

Generalization and Specialization

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Fundamentals

Generalization combines multiple similar entities into a superclass using a bottom-up approach.

Specialization divides a general entity into subclasses using a top-down approach.

Example Hierarchy:

Person

| ISA

Student | Teacher | Administrator

The word ISA means: Student is a Person, Teacher is a Person, Administrator is a Person.

What is ISA?

Definition: ISA (IS-A) Relationship represents an inheritance relationship between a superclass and its subclasses.

Without ISA (Redundancy)

Common attributes are repeated across Student, Teacher, and Administrator (Name, Age, Address).

Problems: Data redundancy, Difficult maintenance, Inconsistent updates.

With ISA (Efficiency)

Common attributes are stored once in a **Person** superclass.

Benefits: Eliminates redundancy, Improves maintainability.

Key Concepts

- **Superclass:** The generalized parent entity that contains attributes and relationships common to all subclasses.
- **Subclass:** A specialized child entity that inherits all common properties of its superclass and may introduce additional attributes.
- **Inheritance:** Means that every subclass automatically receives the attributes, relationships, and Primary Key of its superclass.

Constraints in ISA Hierarchy

Hierarchy:

1. Disjoint / Overlapping
2. Total / Partial Specialization

1. Disjoint vs Overlapping

Disjoint: An entity instance can belong to only one subclass (e.g., Vehicle: Car, Bike, Truck).

Overlapping: An entity instance can belong to multiple subclasses simultaneously (e.g., Person: Student, Teacher).

2. Total vs Partial Specialization

Total Specialization: Every superclass entity must belong to at least one subclass (Double line representation).

Partial Specialization: Some superclass entities may not belong to any subclass.

Database Implementation

Typically using separate tables where the subclass table's primary key is also a foreign key referencing the superclass.

Person	Student	Teacher
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PersonID	PersonID (PK, FK)	PersonID (PK, FK)
Name	RollNumber	Salary
DOB	CGPA	Department

Summary

Generalization emphasizes similarities, while Specialization emphasizes differences. These techniques help eliminate redundancy, improve maintainability, and model real-world hierarchies accurately.